

Author Contributions Statement

Hierarchical Wide-Area Protection Coordination for Multi-Substation IBR Networks:
Delayed Consensus and Scalability Analysis

CRediT taxonomy. **AVE** = Arun V. Eluvathingal; **RKP** = R. K. Prasad (supervisor).

CRediT role	AVE	RKP
Conceptualisation	Lead	Supporting
Methodology	Lead	Supporting
Software	Lead	—
Validation	Lead	Supporting
Formal analysis	Lead	Supporting
Investigation	Lead	—
Resources	—	Lead
Writing – original draft	Lead	—
Writing – review & editing	Lead	Lead
Supervision	—	Lead

Detailed statement.

Arun V. Eluvathingal (AVE): Identified the multi-substation scaling gap in the SAMBPS framework. Designed the three-layer hierarchical architecture and spectral clustering partition. Derived the delayed κ_n -weighted consensus weights and the weighted fault-zone voting rule. Proved Proposition 1 (delayed consensus correctness) using graph-theoretic weight dominance. Implemented the Monte Carlo simulation framework on ANDES (IEEE 39, IEEE 118, synthetic 50-substation), computed all performance metrics. Conducted the scalability and cluster-size sensitivity analyses. Wrote all sections of the manuscript.

R. K. Prasad (RKP): Provided research direction and identified the wide-area scaling gap as a high-priority open problem. Reviewed the consensus algorithm design and the convergence proposition. Provided access to ANDES simulation infrastructure and RTDS interface. Edited and approved the final manuscript.

Plain-text portal version:

AVE: Conceptualisation (Lead); Methodology (Lead); Software (Lead); Validation (Lead); Formal analysis (Lead); Investigation (Lead); Writing – original draft (Lead); Writing – review & editing (Lead).

RKP: Conceptualisation (Supporting); Methodology (Supporting); Validation (Supporting); Formal analysis (Supporting); Resources (Lead); Writing – review & editing (Lead); Supervision (Lead).